

Predictions: Theory and Practice

Part II: Market-based Predictions

Patrick Schmidt

University of Heidelberg

Asset markets as prediction systems

Motivation

Asset prices in news

- daily news routinely reports asset prices
 - “the DAX closed 0.7% higher today”
 - “the Euro lost ground against the Dollar”
 - “Oil prices surged after Putin’s interview...”
 - “OpenAI evaluated at 900bn dollars after...”
- but most listeners do not actually trade these assets
- they are treated as informative about the state of the world

- **market prices tell us something about the future**
- we cover
 - which property of the future is being predicted?
 - how reliable is that prediction?

Which asset prices?

A few examples we will keep in mind:

- **stock price:** claim on future dividends (and resale value of a company)
- **exchange rate:** relative value of two currencies (or the basket of goods one can buy with it)
- **commodity futures:** contract for delivery of oil/wheat/... at a fixed future date
- **bond price:** claim on future interest payments

common structure:

- a price S_t observable today
- a future payoff Y (dividends, exchange rate, delivery price, ...)
- price S_t is known
- Y is uncertain $\Rightarrow$ predict with probability distribution
- our question: which property of Y does S_t predict?

Market prices as mean predictions

From traders' beliefs to market prices

- consider an asset with payoff Y at $t + 1$, traded at price S_t today
- assume for now: no time value of money, risk-neutral traders (no risk premium), homogeneous beliefs
- a trader with belief $E[Y]$ about the payoff
 - buys if $E[Y] > S_t$
 - sells if $E[Y] < S_t$
- market clears when supply equals demand

$$\Rightarrow S_t = E[Y]$$

- the market price equals the expected payoff

...modulo some issues

In reality, three things complicate the simple picture $S_t = E[Y]$:

1. **discounting**: money tomorrow $\neq$ money today

$$S_t = \frac{E[Y]}{1 + r}$$

2. **risk premia**: investors are not risk neutral

$$S_t = \frac{E[Y]}{1 + r + \pi}$$

where π is a compensation for bearing risk

3. **heterogeneity in beliefs**: Traders $i = 1, \dots$ may have different mean predictions $E_i[Y]$

$$S_t = \text{median of } E_i[Y]$$

(or wealth weighted if heterogeneous wealth of traders)

- takeaway: price tracks **risk-adjusted** and **wealth-weighted** mean predictions

The Efficient Market Hypothesis

Definition

Efficient Market Hypothesis (EMH)

A market is **efficient with respect to an information set $\mathcal{I}$** if asset prices fully reflect that information:

$$S_t = E[Y \mid \mathcal{I}] / (1 + r + \pi)$$

- equivalently: no trading strategy based on $\mathcal{I}$ earns systematic excess returns
- the “fair game” property: returns conditional on $\mathcal{I}$ are unpredictable beyond the risk-adjusted required return
- different choices of $\mathcal{I}$ give the three forms on the next slide

Three forms of EMH

(Fama 1970) distinguishes three forms by the information $\mathcal{I}$ that prices reflect:

- **weak form:** $\mathcal{I}$ = past prices
 - implication: returns are unforecastable from past returns
- **semi-strong form:** $\mathcal{I}$ = all public information
 - implication: news is incorporated in prices immediately
- **strong form:** $\mathcal{I}$ = all information (incl. private)
 - implication: even insiders cannot earn excess returns

most empirical work focuses on weak and semi-strong forms.

The random walk model

- consider a simple one-one period asset model:
 - hold today, sell tomorrow, so the payoff is $Y = S_{t+1}$
 - no risk aversion/premium $\pi = 0$
 - no interest/time value of money $r = 0$
- EMH:

$$S_t = E[S_{t+1} \mid \mathcal{I}_t]$$

- EMH:

$$S_t = E[S_{t+1} \mid \mathcal{I}_t]$$

- implies unpredictable errors

$$E[S_{t+1} - S_t \mid \mathcal{I}_t] = 0$$

- define returns $\varepsilon_{t+1} := S_{t+1} - S_t$
- returns are unpredictable:

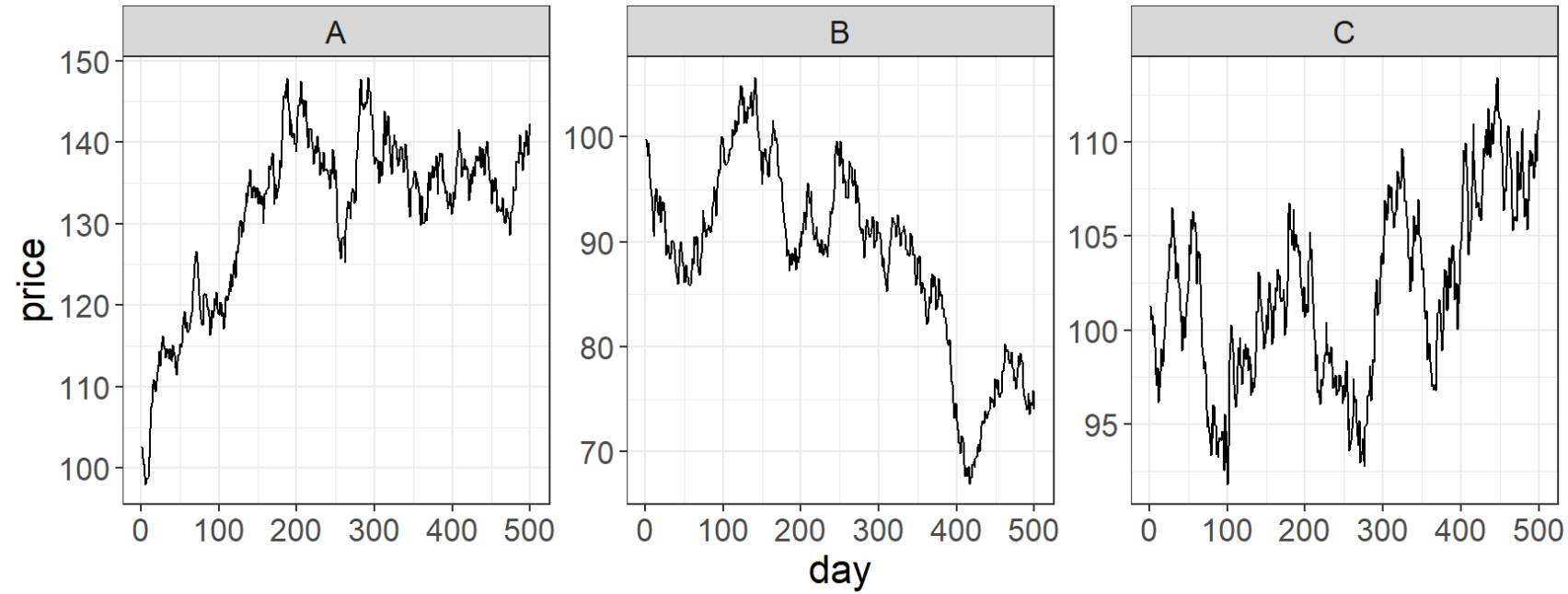
$$E[\varepsilon_{t+1} \mid \mathcal{I}_t] = 0$$

- The EMH implies the **random walk** model for prices

$$S_{t+1} = S_t + \varepsilon_{t+1}, \quad E[\varepsilon_{t+1} \mid \mathcal{I}_t] = 0$$

- best forecast of tomorrow's price = today's price
- anything that *could* be forecast from $\mathcal{I}_t$ is already in S_t
- with non-zero r and π : similar results hold for **excess returns** $r_{t+1} - (r + \pi)$

Looks like a stock chart?



- all three are simulated random walks
- visually indistinguishable from real stock charts
- humans are very good at seeing patterns where there are none
- Efficient market hypothesis implies:
 - Active stock picking and market timing are generally futile
 - Theoretical basis for passive index funds

Empirical evidence

Evidence supportive of EMH

- **active managers underperform**
 - most actively managed equity funds are beaten by passive index benchmarks (after fees)
- **event studies**
 - prices react to earnings, M&A, macro news within minutes
 - post-news returns close to zero on average
- **exchange rates: the Meese-Rogoff puzzle (Meese and Rogoff 1983)**
 - structural macro models cannot beat the random walk in out-of-sample forecasts

Anomalies against EMH

- **post-earnings announcement drift (PEAD)** (Bernard and Thomas 1989)
 - after a positive earnings surprise, returns continue drifting up for 60-90 days
- **momentum** (Jegadeesh and Titman 1993)
 - past 3-12 month winners keep outperforming past losers
- **excess volatility** (Shiller 1981)
 - aggregate stock prices move much more than changes in expected dividends justify

How to read anomalies

- **joint hypothesis:** Two interpretations of any anomaly:
 - i. market is **inefficient** → genuine arbitrage opportunity
 - ii. asset pricing **model is incomplete** → extra return is compensation for a risk factor we failed to measure
- if arbitrage possible, market will probably correct after publication
 - investors will seek out excess returns, price increases, excess vanishes
- Second option: **inference and multiple testing**
 - anomalies from EMH look like clear statistical results
 - but if many projects study different anomalies, we expect 1 in 10 to be statistically significant at 0.1 α -level even if hypothesis true
 - if only this one study is published/catches our attention, inference is flawed

Why market prices still predict well

even granting the anomalies, market prices remain remarkably hard to beat as predictions:

- **selection**
 - persistently bad traders lose money and exit
 - the surviving population is at least somewhat accurate
- **incentives**
 - capital at risk is a much stronger incentive than answering a survey
 - skin in the game (cf. discussion of judgment forecasts)
- **information aggregation**
 - many participants, diverse private signals
 - prices summarise far more information than any single forecaster has

Beyond the mean: market-implied uncertainty

Markets price more than the mean

- so far: market price $S_t \approx E[Y]$
 - market gives us a **point prediction** of the future payoff
- but predictions ideally come with uncertainty
- can markets also reveal **distributional** information?
 - yes with **options**

Aside: Odd Lots on the VIX

- Bloomberg's *Odd Lots* podcast
 - Episode: “What Just Happened in Markets? Volatility and the Big Selloff”
 - on Spotify etc.

What is an option?

- a financial contract whose payoff depends on the future price S_T of an underlying asset
- **European call option** with strike K , expiry T :
 - right (not obligation) to **buy** the underlying at prize K at time T
 - payoff at T : $\max(S_T - K, 0)$
- **European put option** with strike K , expiry T :
 - right to **sell** the underlying for prize K at time T
 - payoff at T : $\max(K - S_T, 0)$
- actively traded on exchanges (CBOE, Eurex, Eurex...)

From option prices to implied distributions

- how can we read an implied probability P directly from call prices?
- consider a **call spread**: buy call at K , sell call at $K + \Delta K$

$$\text{payoff} = \begin{cases} \Delta K & \text{if } S_T > K + \Delta K \\ S_T - K & \text{if } K < S_T \leq K + \Delta K \\ 0 & \text{if } S_T \leq K \end{cases}$$

- for small ΔK : $\text{payoff} \approx \Delta K \cdot \mathbf{1}\{S_T > K\}$

- no-arbitrage pricing: costs = discounted expected payoff

$$C(K) - C(K + \Delta K) \approx e^{-rT} \cdot E[\Delta K \cdot \mathbf{1}(S_T > K)]$$

$$e^{rT} \frac{C(K) - C(K + \Delta K)}{\Delta K} \approx P(S_T > K)$$

- taking $\Delta K \rightarrow 0$ and using $P(S_T \leq K) = 1 - P(S_T > K)$:

$$\boxed{1 + e^{rT} \frac{\partial C}{\partial K} = P(S_T \leq K)}$$

- **the CDF of S_T** is directly readable from the slope of the call-price curve
- so option markets give us not just a mean, but an entire **predictive distribution**

Aside: Similar result for density

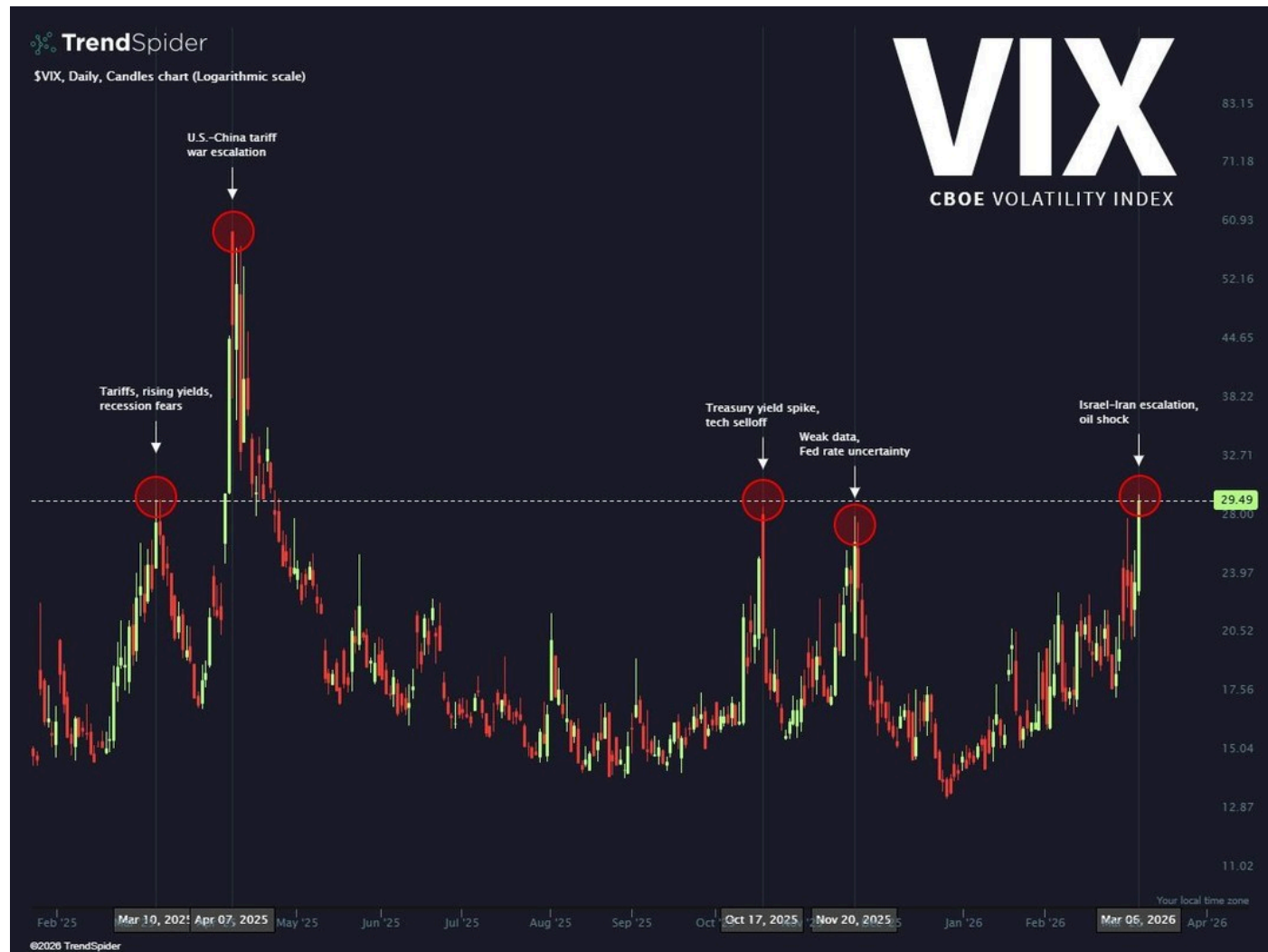
Breeden–Litzenberger (1978)

The second derivative of the call price $C(K)$ with respect to strike K recovers the (risk-neutral) density of S_T :

$$f_{S_T}(K) = e^{rT} \frac{\partial^2 C(K)}{\partial K^2}$$

The VIX: a single number for uncertainty

- Cboe Volatility Index (VIX)
 - launched 1993, current methodology since 2003
 - aggregates S&P 500 option prices into one number
 - reports market-implied 30-day annualised volatility of the S&P 500
- often called the “**fear gauge**”
 - VIX spikes in crises (e.g., financial crisis 2008, Covid 2020)
 - quoted on daily news as a macro uncertainty indicator
- has become an asset class in its own right
 - VIX futures, options on VIX



VIX daily.

VIX methodology (sketch)

- realised variance over $[0, T]$ with n daily log-returns $r_i = \log(S_i/S_{i-1})$:

$$RV_T = \frac{1}{T} \sum_{t=1}^T r_t^2$$

- the price should be

$$E[e^{rT} RV_T]$$

- VIX is the market's risk-neutral expectation of 30-day realised variance
- it can be replicated by a portfolio of calls and puts

VIX as a prediction

- the VIX is a (risk-adjusted) **mean prediction of realised volatility** over the next 30 days
 - target Y = annualised std of S&P 500 returns over $[t, t + 30]$
- empirical record (**Bollerslev et al. 2009**)
 - VIX and subsequent realised vol are highly correlated ($\rho \approx 0.86$)
 - Thus, option prices predict uncertainty successfully
 - VIX is on average **higher** than the subsequent realised vol
 - Thus, investors pay a premium for protection against volatility spikes

Wrap-up

Take-aways

- in an efficient market, the price is a (risk-adjusted) **mean prediction** of future payoffs
- **changes in prices** are unpredictable
 - basis for the random walk model
- empirically:
 - hard to beat market prices on average (exchange rates, fund performance)
 - some documented anomalies
 - interpretation of anomalies limited by the **joint hypothesis problem** and **multiple testing**
- option markets extend this: prices reveal the **distribution** of future outcomes
 - VIX as a market-based uncertainty forecast
- the next section turns to **prediction markets**
 - artificial markets specifically designed to reveal predictions

Prediction Markets

Prediction markets

- prediction markets ([Arrow et al. 2008](#))
 - artificial markets with main purpose of revealing information
 - idea: Trade contracts on free market
 - contract pays depending on target variable y
- contrary to standard market prices
 - also reveal information
 - but their primary use/goal is allocation

Aside:



- Last week tonight by John Oliver on Prediction markets

Example prediction market contract

- contract pays 1\$ if event happens
- what is its equilibrium price?
- if you belief probability is p you buy if

$$\begin{aligned} E_p[\text{money if buy}] &> E_p[\text{money if not buying}] \\ p \times 1 + (1 - p) \times 0 &> \text{price} \\ p &> \text{price} \end{aligned}$$

- market in equilibrium if demand and supply of contracts coincide
 - need as much buyers as sellers of contract
- Thus, equilibrium price connected to beliefs p in market

- “Price in prediction market represents market belief”
 - vague statement that is generally not true
- Can be derived under simplifying assumptions about market participants
 - risk neutral
 - identical wealth
 - heterogenous beliefs p
 - then: market equilibrium price = median(p)
- Fails under more realistic assumptions
 - risk averse: participation less likely
 - utility correlated with outcome: hedging
 - homogenous beliefs: no liquidity on market
 - wealth differences: weighted median

Calculating the margin

- traditional bookmakers quote **odds**, not probabilities
 - decimal odds o : a 1\$ stake returns o \$ if you win
 - implied probability of a single outcome: $\hat{p} = 1/o$
- for a fair market the implied probabilities sum to 1
- bookmakers shade the odds so they sum to **more than 1**:

$$\text{margin} = \left(\sum_k \frac{1}{o_k} \right) - 1$$

- the excess is the bookmaker's expected profit
- example: two equally likely outcomes quoted at $o = 1.91$
 - implied prob each: $1/1.91 \approx 0.5236$
 - sum $\approx 1.0472 \Rightarrow$ margin $\approx 4.7\%$

Prediction markets won't do Bayesian updating

- nice example from Ottoviani et al.(2015) on updating in prediction markets:
 - hypothetical market based on which team, Italy or Denmark, will win a soccer game
 - traders who are subjectively more optimistic about Italy winning live further south
 - traders living south of a certain threshold latitude invest all their wealth in the asset that pays if Italy wins
 - likewise, traders north of the threshold latitude invest all their wealth in the Denmark asset

- what happens when traders observe information (such as a player injury) more in favor of Italy winning?
 - This information causes the price of the Italy asset to be higher, while contemporaneously reducing the price of the Denmark asset
 - the southern traders (who are optimistic about Italy) are able to buy fewer Italy assets
 - northern traders can afford, and thus demand, more Denmark assets, now cheaper.
 - for the market to equilibrate, some northern traders must turn to the Italian side
- Thus, when information more favorable to an outcome is available, the marginal trader who determines the price has a prior belief that is less favorable to that outcome
 - market adjustment and heterogeneity in priors reduces the effect of information on equilibrium price

Empirical evidence

- but empirical evidence for prediction markets rather positive
- today routinely covered in media as reference point
 - arguably more representative than experts
- real advantage probably related to selection into participation
 - less accurate market participants unlikely to stay in market
 - incentives for market participants to get more accurate

Information and Insider trading

The Grossman-Stiglitz paradox

(Grossman and Stiglitz 1980) point out a tension at the heart of the EMH:

- gathering information is **costly** (research, data, analysis)
- a trader only pays this cost if it lets them earn excess returns
- but if prices **already fully reflect all information**:
 - the informed trader has no advantage
 - nobody bothers to gather information
 - then prices reflect **nothing**
- $\Rightarrow$ a perfectly efficient market is **impossible** (self-contradictory)

Resolution: partial efficiency

- markets settle at an equilibrium that is efficient *enough*, not perfectly
- prices must stay **slightly mispriced** so that:
 - informed traders earn just enough to cover their information costs
 - this is the “reward” for producing information
- requires some **noise** (uninformed / liquidity traders)
 - otherwise informed trades instantly reveal the signal and the edge vanishes
- takeaway: a little inefficiency is **necessary**

Insider trading in normal markets

- in stock/asset markets, **insider trading is illegal**
 - trading on material, non-public information
- why ban something that makes prices more informative?
- main worry: **market liquidity**
 - market makers and uninformed traders systematically lose to insiders
 - this is **adverse selection**: ask for a trade, and you only get a trade when an insider knows better
 - response: less trades, wider spreads, or even stop of trading
- a market dominated by insiders **dries up**
 - the “lemons” problem of separated market equilibria for liquidity

Insider trading in prediction markets

- prediction markets exist **precisely to aggregate private information**
- but the people with private information *are* the insiders!
 - the employee who knows the product will ship late
 - the staffer who knows the poll internals
- **direct conflict**: banning insider trading blocks exactly the information the market is designed to reveal
- the insider's “unfair advantage” is the **valuable signal** we want in the price
- open question how this develops

References

- Arrow, Kenneth J, Robert Forsythe, Michael Gorham, et al. 2008. “The Promise of Prediction Markets.” In *Science*, No. 5878, vol. 320. American Association for the Advancement of Science.
- Bernard, Victor L, and Jacob K Thomas. 1989. “Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?” *Journal of Accounting Research* 27: 1–36.
- Bollerslev, Tim, George Tauchen, and Hao Zhou. 2009. “Expected Stock Returns and Variance Risk Premia.” *Review of Financial Studies* 22 (11): 4463–92.
- Fama, Eugene F. 1970. “Efficient Capital Markets: A Review of Theory and Empirical Work.” *Journal of Finance* 25 (2): 383–417.
- Grossman, Sanford J, and Joseph E Stiglitz. 1980. “On the Impossibility of Informationally Efficient Markets.” *The American Economic Review* 70 (3): 393–408.
- Jegadeesh, Narasimhan, and Sheridan Titman. 1993. “Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency.” *Journal of Finance* 48 (1): 65–91.
- Meese, Richard A, and Kenneth Rogoff. 1983. “Empirical Exchange Rate Models of the Seventies: Do They Fit Out of Sample?” *Journal of International Economics* 14 (1–2): 3–24.
- Ottaviani, Marco, and Peter Norman Sørensen. 2015. “Price Reaction to Information with Heterogeneous Beliefs and Wealth Effects: Underreaction, Momentum, and Reversal.” *American Economic Review* 105 (1): 01–34.
- Shiller, Robert J. 1981. “Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?” *American Economic Review* 71 (3): 421–36.